

Choose a set of data science concepts relevant to your area of interest and better also be related to your final project as much as possible. Use the material to create an ontology using Protege and/or Gephi. Prepare and document the competency questions list of 6. Create the agent, share it within the group. Assign group members responsible for a section of the competency questions so they can create their test-cases. Start training the chatbot agent as a group. Test each other sections quality, going through the test-cases.

Your development is online using Dialog-Flow, Google; and no coding is necessarily needed for this assignment. Test-cases should be documented based on an expansion of competency questions; create 5 different ways of asking each competency question. All training and evaluations are done dynamically online. Prepare communications so another group can test your chat-bot agent. Share your intents with another group.

1- Prepare: to choose the Data Science subjects that you'd want to enable the chatbot with. I mean, just as a suggestion you can choose from the topics of your Data Science course that you'd want to enable the chatbot to learn about and answer, but the context can be in any topic of your choice.

2- Create the ontology: use Protege to develop the knowledge structure.

3- Competency Questions: choose the atomistic competency questions.

4- Test-Cases: Design your test questions and answers; i.e. alternative ways of asking competency questions. 5- Training Question-Answers: Intents Development using Google Dialog Flow.

6- Evaluate, measure the quality to report.

Document your steps, test-cases, explain the results effectively, plus Protege file. Present, by letting user Verify and validate and Evaluate.

Your deliverable should include: Ontology, Competency questions and test-cases and the chatbot instance to be shared with your TA to evaluate.

Rubric: (accounts for 10% of the final grade.)

Create the ontology of the chosen material 3%

Competency Questions and Test-Cases 2%

Training Question-Answers, Intents Development 2%

Evaluate to isolate weaknesses 2%

Reports, Presentation 1%